

Statistical tests

how they work

a talk by Pavel Popov

Introduction

- I know some probability, and I use statistical tests to validate my findings
- But I am at a loss as to how they work:
 - Do I need one- or two-tailed test? paired or unpaired? equal variance? rank test? 🤪
- So I want to rebuild a simple test from scratch and figure it out in the process
- Then I'll also talk about a paper

1. Foundation stuff

Setting: $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} P, \quad \bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$

$$\mu = \mathbb{E}_P[X_i], \quad \sigma^2 = \mathbb{E}_P[(X_i - \mu)^2]$$

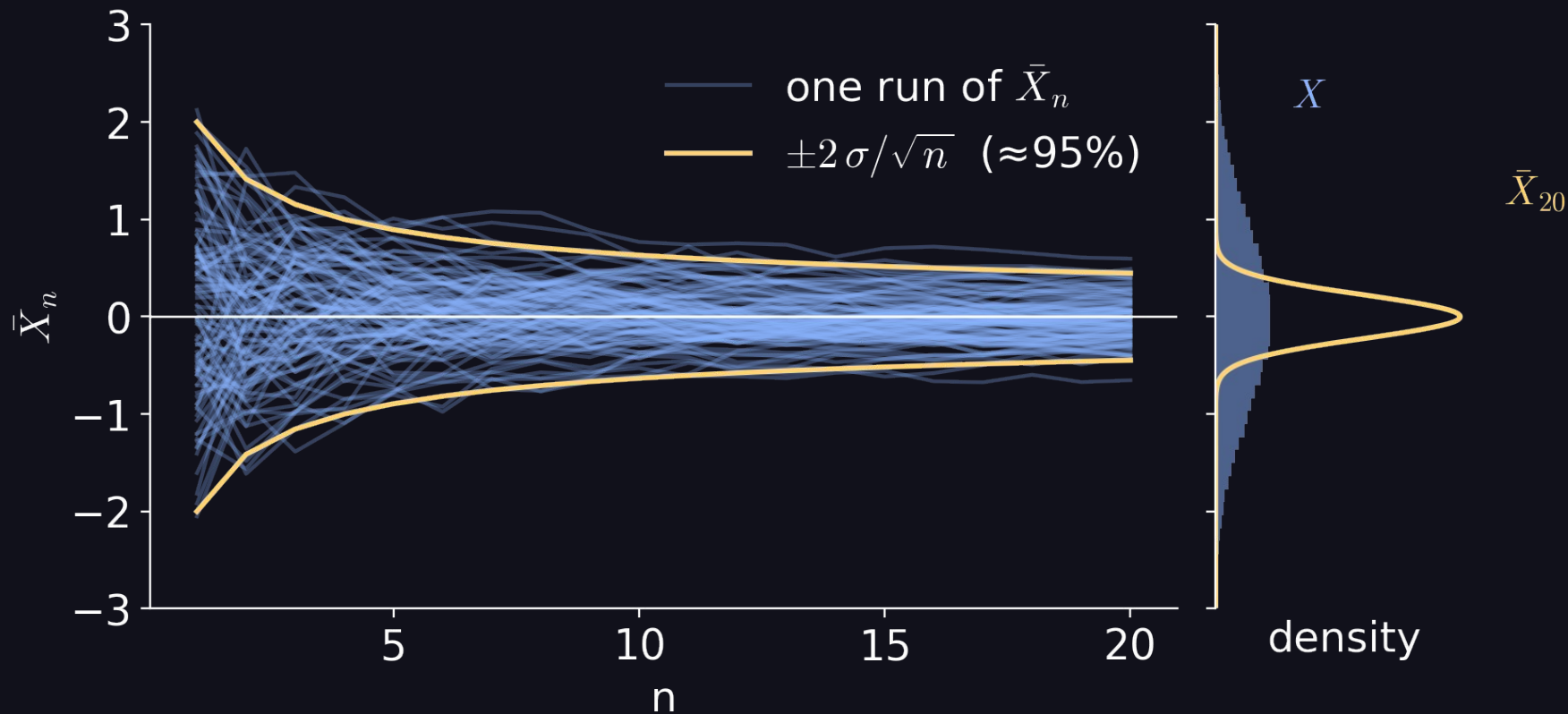
Law of large numbers (LLN): $\bar{X}_n \xrightarrow[n \rightarrow \infty]{\text{a.s.}} \mu$

Central limit theorem (CLT): $\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1)$

1. Two CLT formulations

$$\frac{\bar{X}_n - \mu}{\sigma / \sqrt{n}} \xrightarrow{d} N(0, 1) \quad \Longleftrightarrow \quad \bar{X}_n \xrightarrow{d} N\left(\mu, \frac{\sigma^2}{n}\right)$$

1. Visualization of LLN and CLT



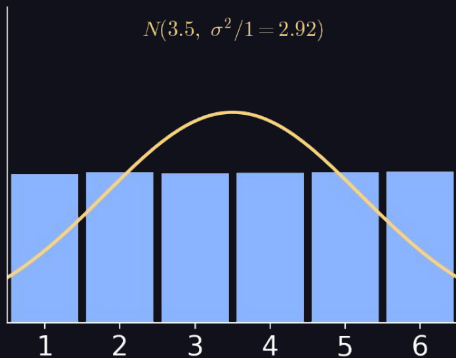
1. CLT applied to different distributions

Uniform distribution (dice)

Dice example

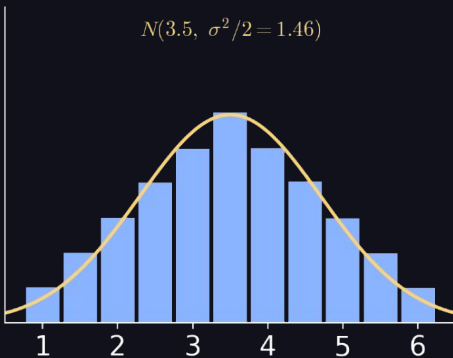
$n = 1$

$N(3.5, \sigma^2/1 = 2.92)$



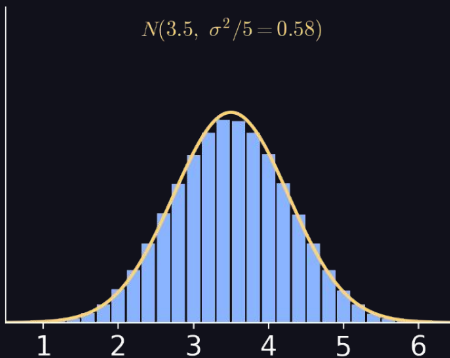
$n = 2$

$N(3.5, \sigma^2/2 = 1.46)$



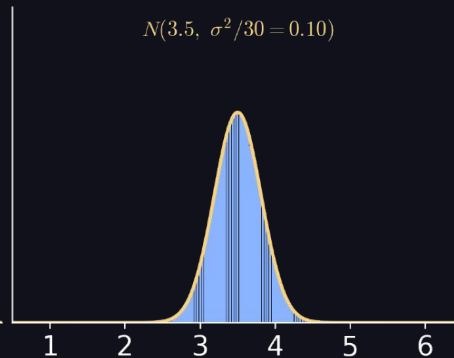
$n = 5$

$N(3.5, \sigma^2/5 = 0.58)$



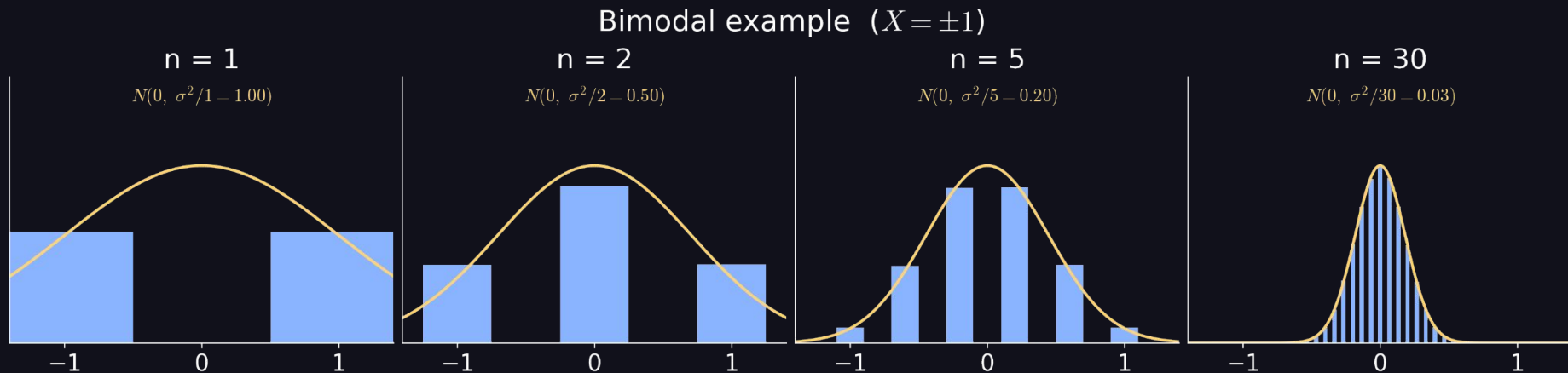
$n = 30$

$N(3.5, \sigma^2/30 = 0.10)$



1. CLT applied to different distributions

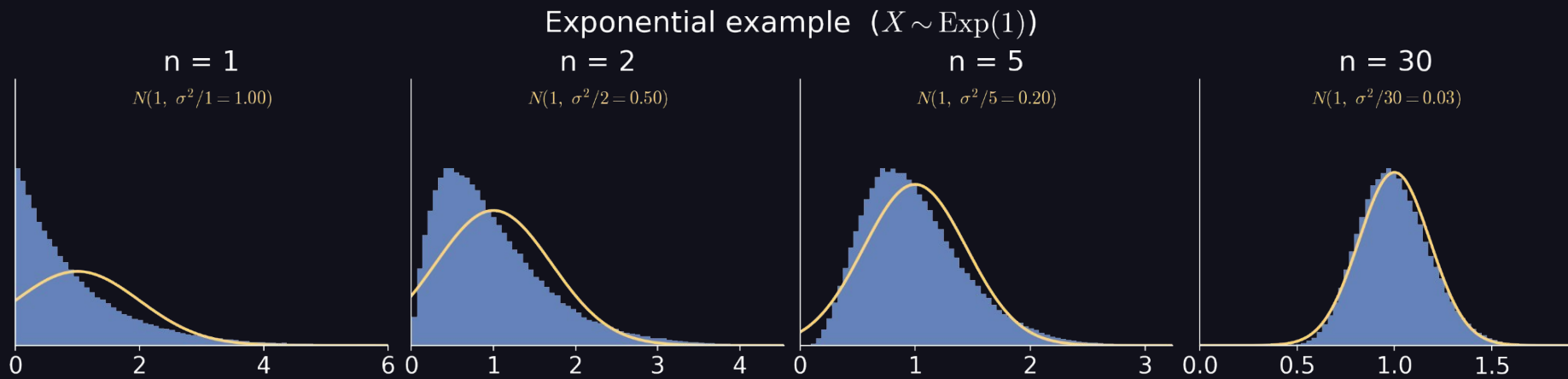
Bimodal distribution (coin flip)



1. CLT applied to different distributions

Skewed distribution (exp)

$$\text{Exp}(1) = e^{-x}$$



1. Limits of CLT: undefined mean and variance

$$\text{Cauchy}(0, 1) = \frac{1}{\pi(1+x^2)}$$

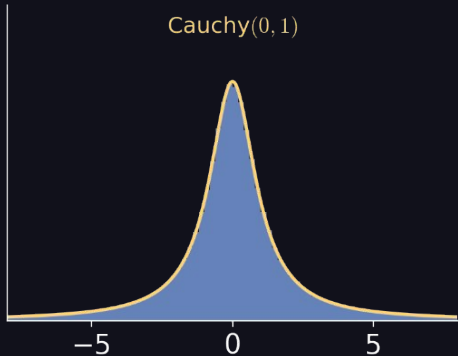
$$X_i \sim \text{Cauchy}(0, 1) \implies \bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \sim \text{Cauchy}(0, 1) \quad \forall n$$

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} \frac{x}{\pi(1+x^2)} dx = \left[\frac{\ln(1+x^2)}{2\pi} \right]_{-\infty}^{\infty} = “\infty - \infty” \implies \mu \text{ undefined}$$

Cauchy example (infinite variance)

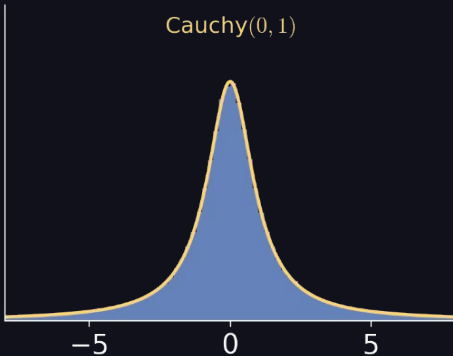
n = 1

Cauchy(0, 1)



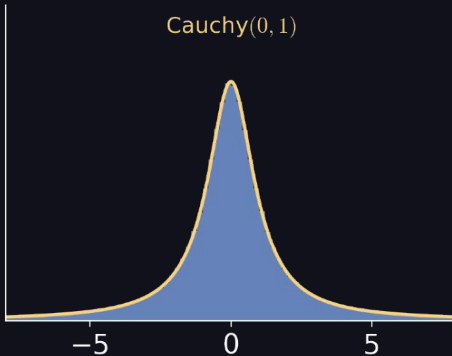
n = 2

Cauchy(0, 1)



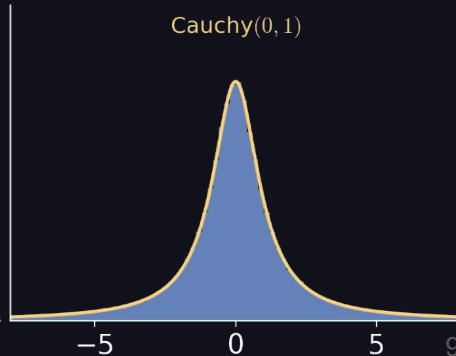
n = 5

Cauchy(0, 1)

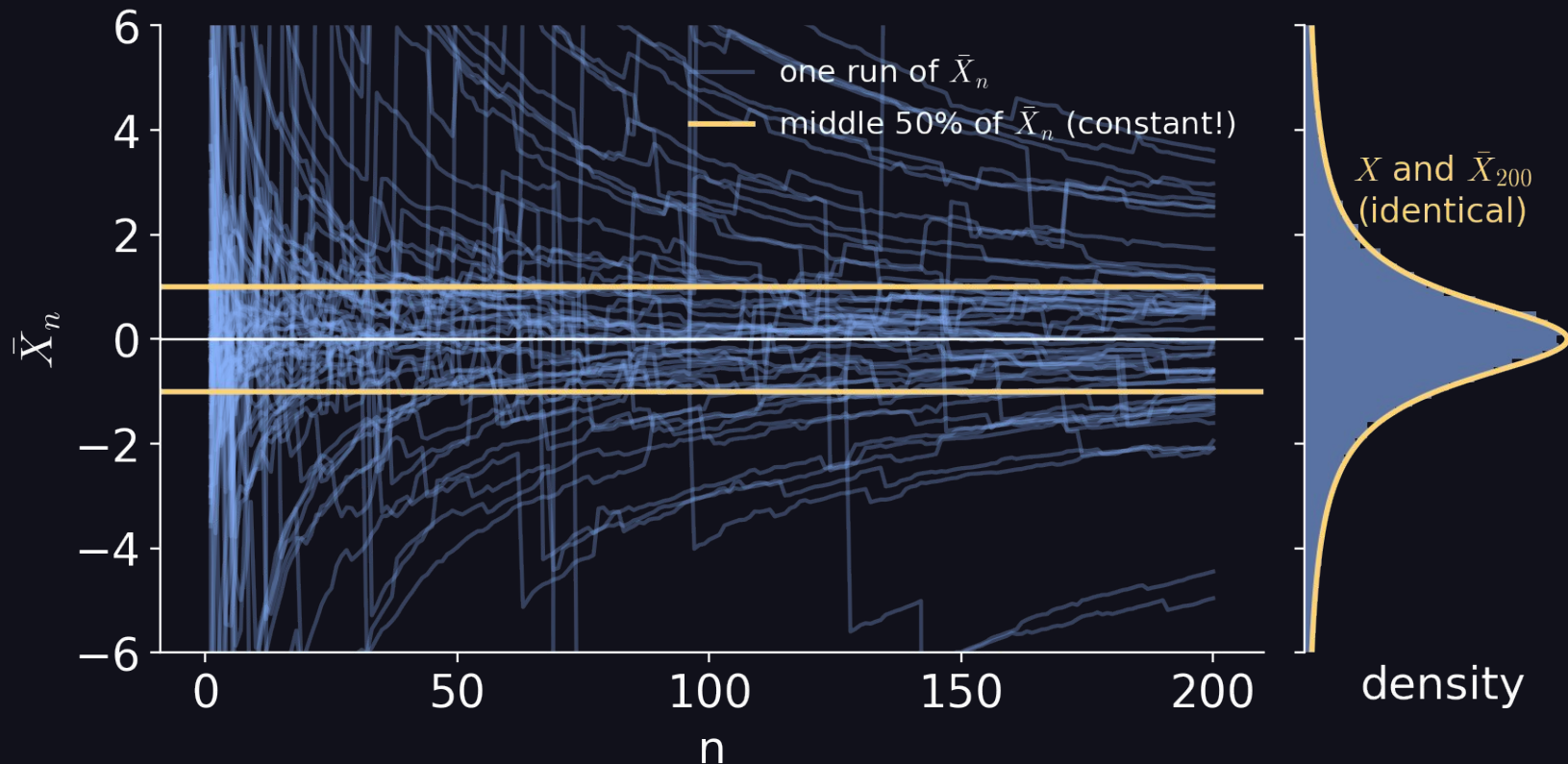


n = 300

Cauchy(0, 1)



1. Cauchy breaks LLN too



1. CLT manifests in deep learning

Neural pre-activations are sums of random things

$$z = \sum_{i=1}^n w_i x_i \xrightarrow{\text{CLT}} z \approx N\left(0, \sigma_w^2 \sum_i x_i^2\right), \quad w_i \stackrel{\text{iid}}{\sim} (0, \sigma_w^2)$$

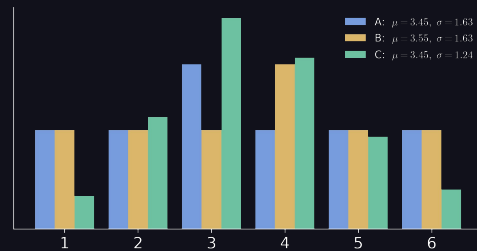
The variance gets out of hand with many dimensions

$$\text{Var}(z) \approx n \sigma_w^2 \overline{x^2} \implies \sigma_w^2 = \frac{1}{n} \text{ (Xavier)}, \quad \frac{2}{n} \text{ (He)}$$

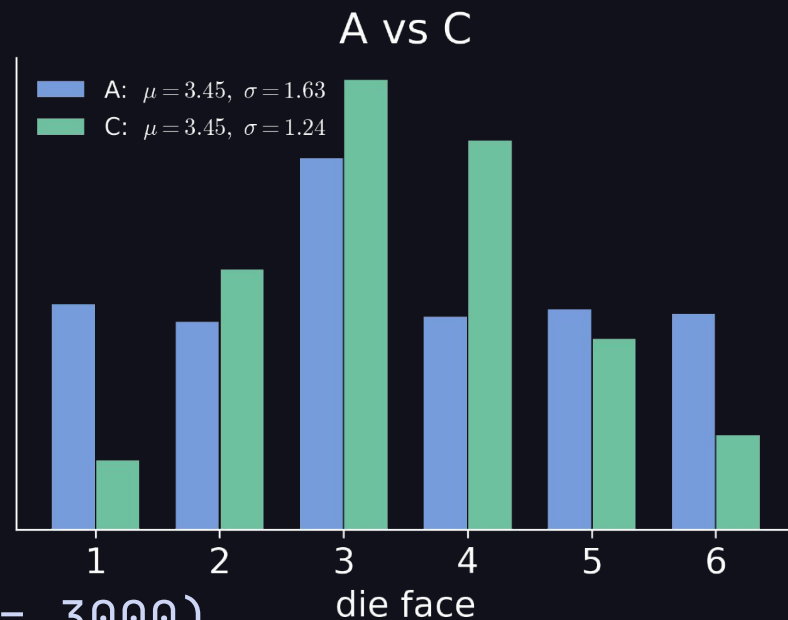
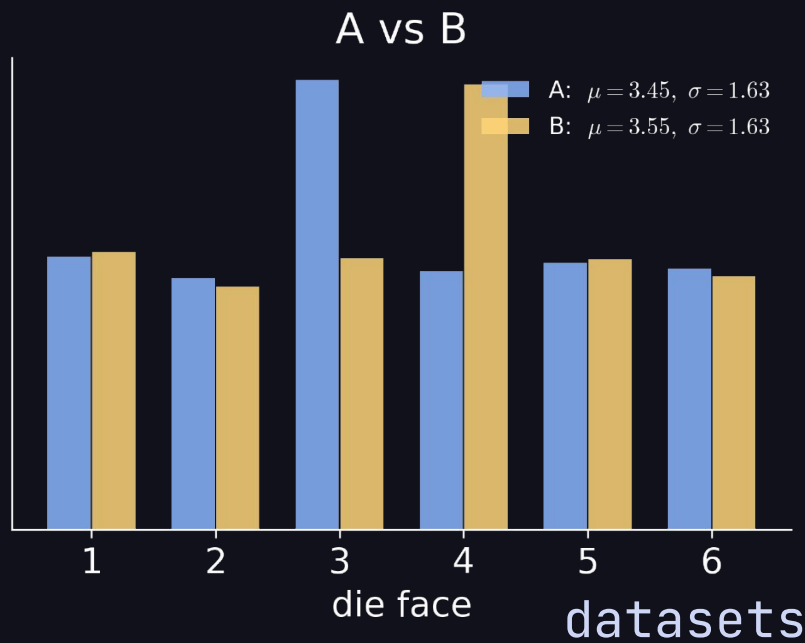
1. How CLT is useful for stat inference?

- Imagine that we have two sets of outcomes, and we want to find out if they are significantly different.
- **We use the CLT's Gaussian to get probability estimates on how likely the difference in outcomes is under pure chance.**
- Making this Gaussian test what we want is where the zoo of statistical tests comes from.

2. Building a statistical test

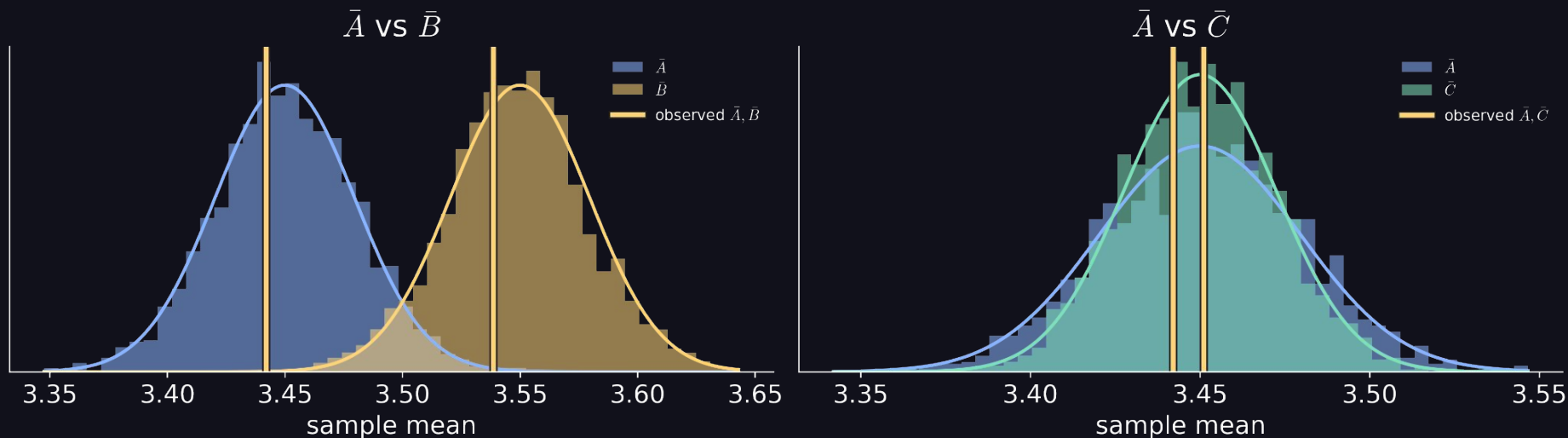


Let's assume we have datasets on 3 dice: A, B, C



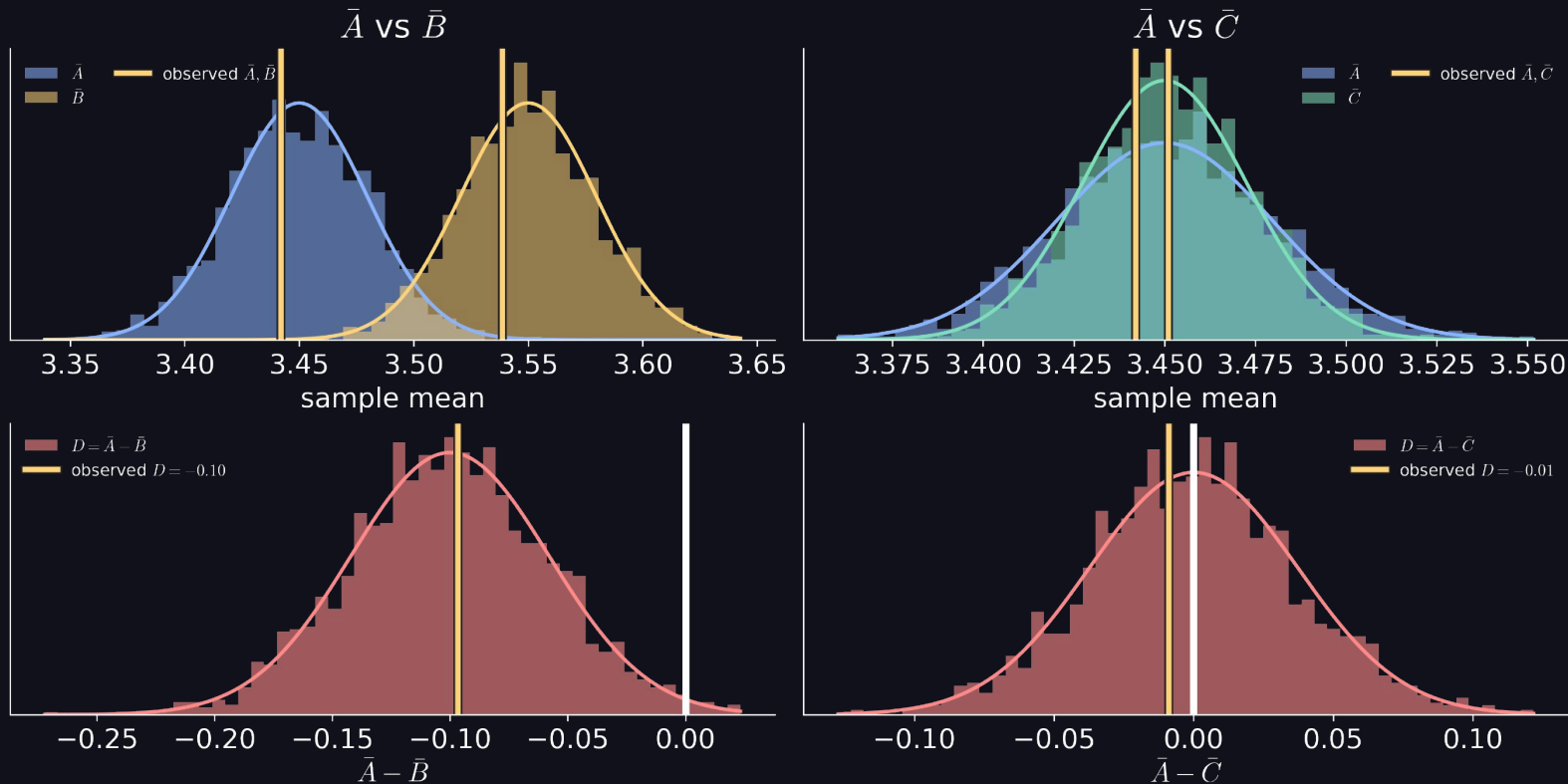
2. Going to sample means

Sample means are gaussians according to CLT

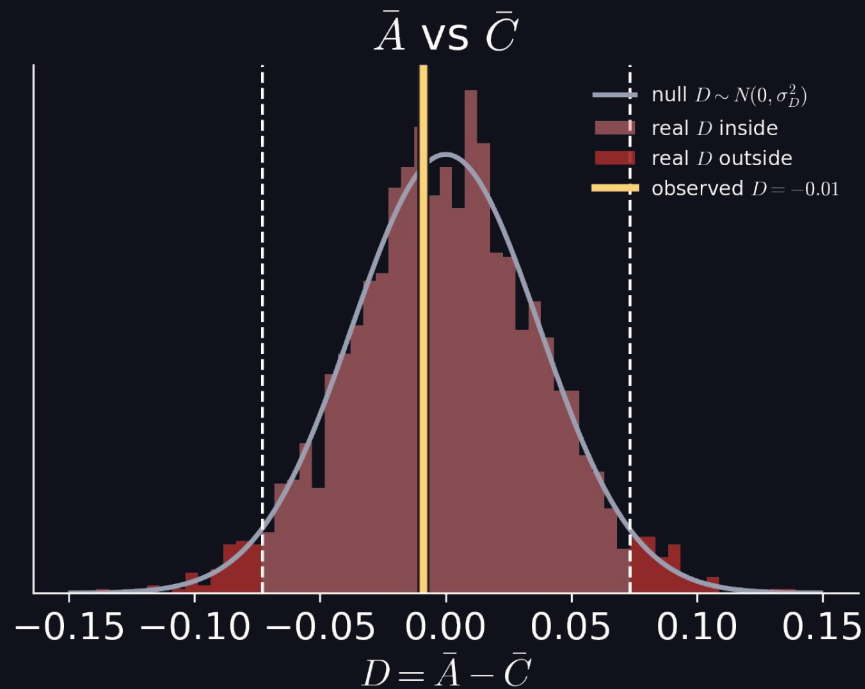
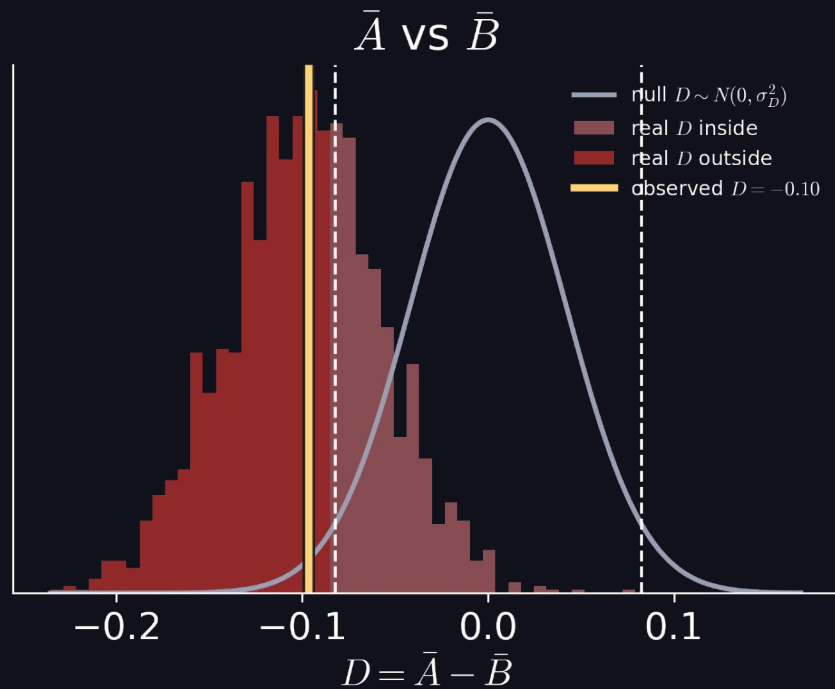


2. Introducing D

$$D = \bar{A} - \bar{B} \sim N\left(\mu_A - \mu_B, \frac{\sigma_A^2}{n_A} + \frac{\sigma_B^2}{n_B}\right)$$

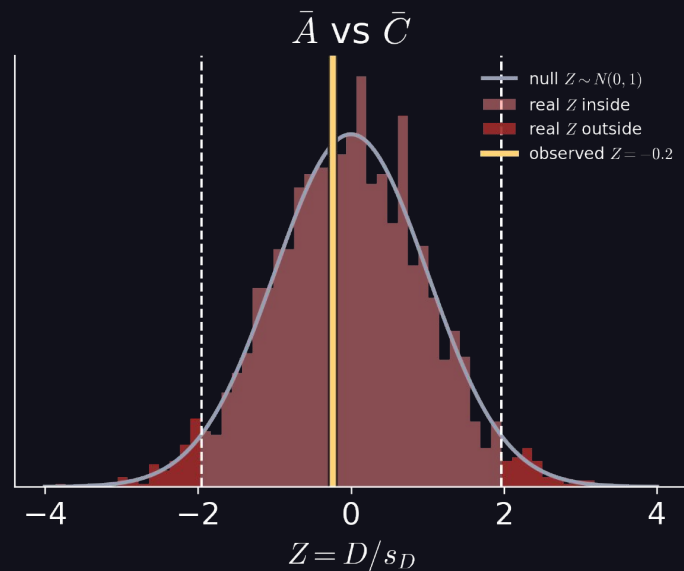
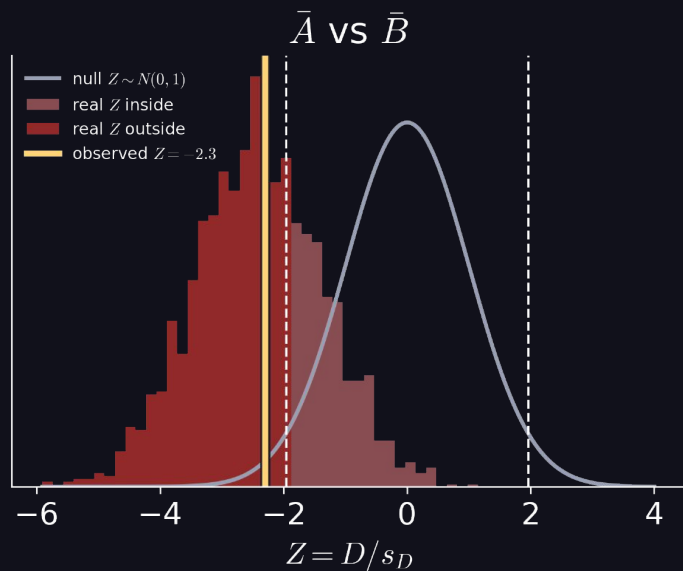


2. Null hypothesis – means are the same

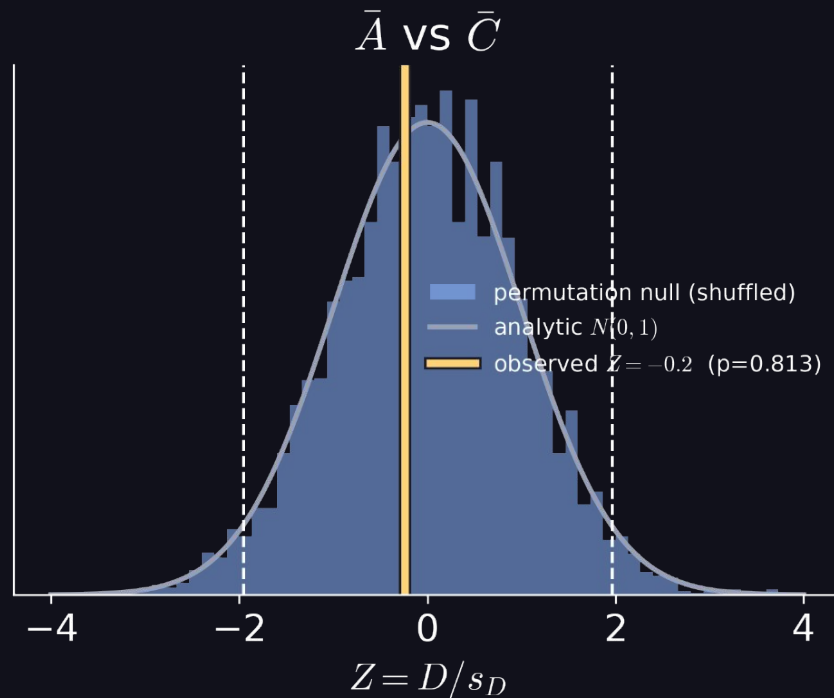
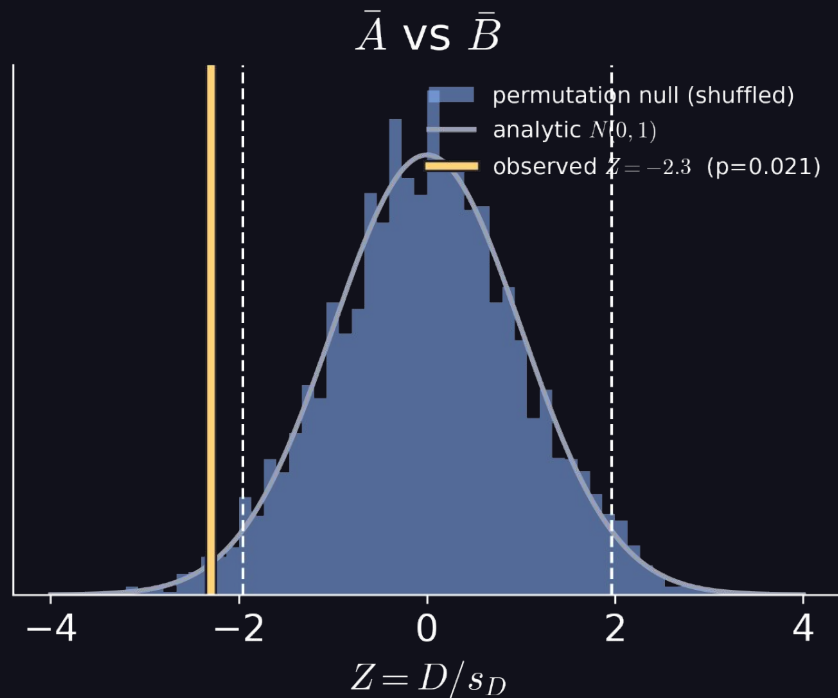


2. Normalize by variance for convenience

$$Z = \frac{\bar{A} - \bar{B}}{\sqrt{s_A^2/n_A + s_B^2/n_B}} = \frac{D}{s_D} \stackrel{H_0}{\sim} N(0, 1)$$

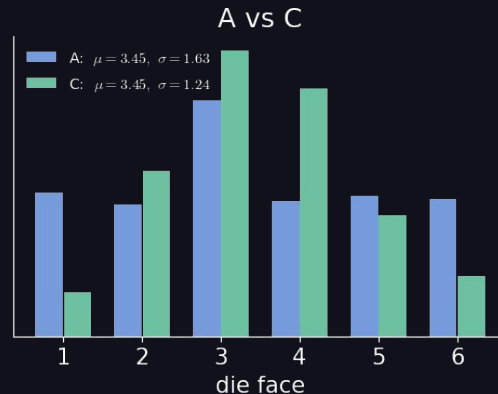
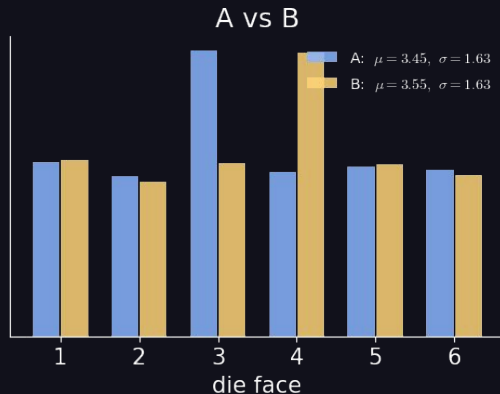


2. Finishing touch – permutation on observed data

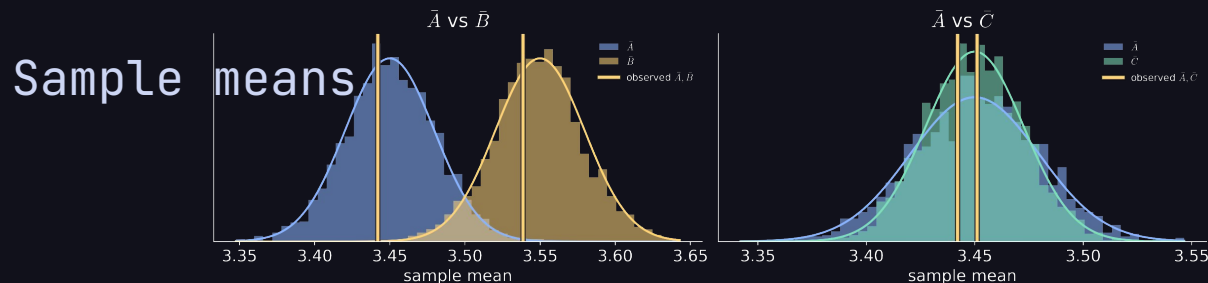


2. What does it mean?

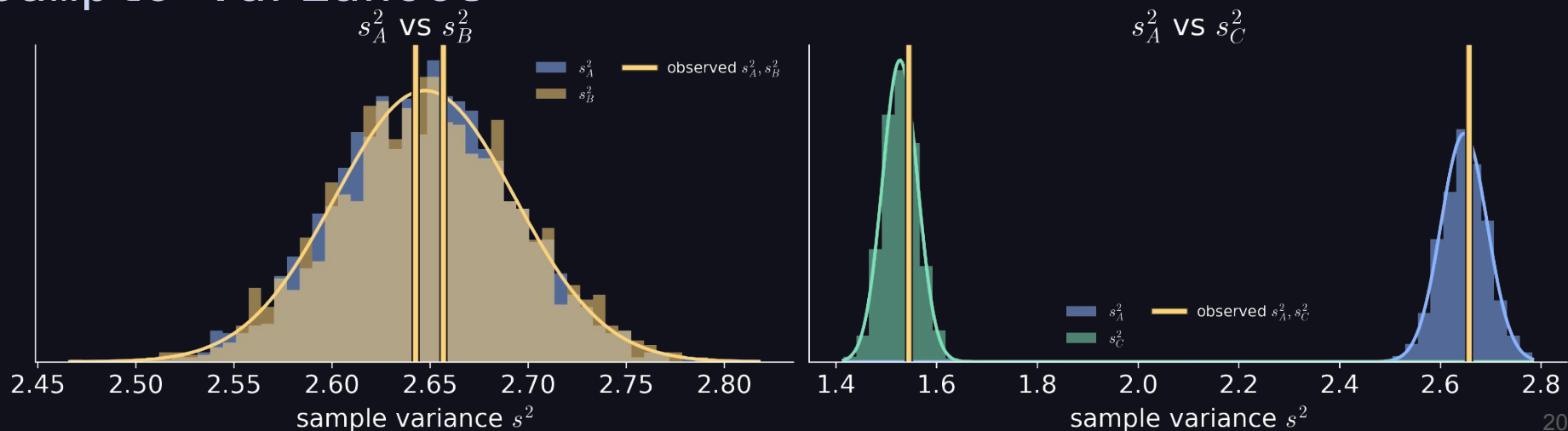
- We've just (de)constructed a permutation test that determines if 2 datasets have the same mean
- So it distinguished A and B, but not A and C.
- How to distinguish A and C?



2. Another test that uses sample variances

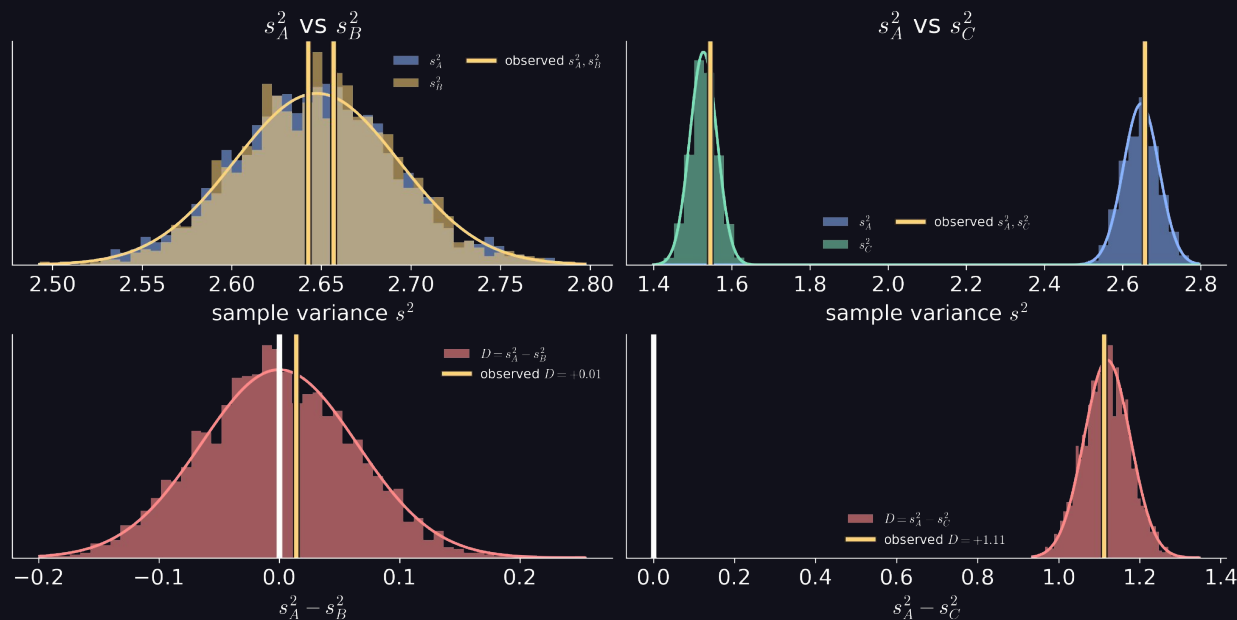


Sample variances

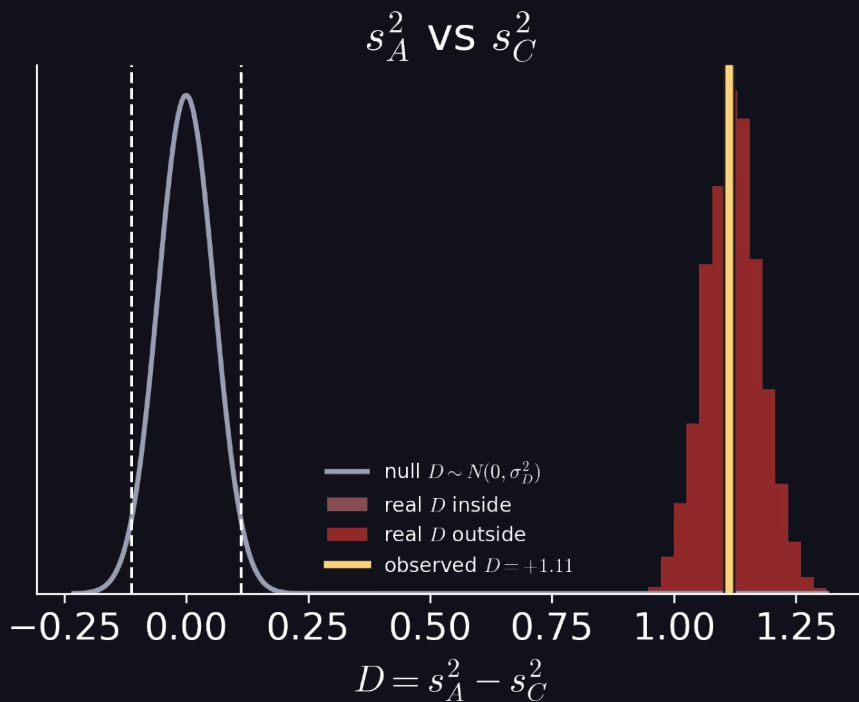
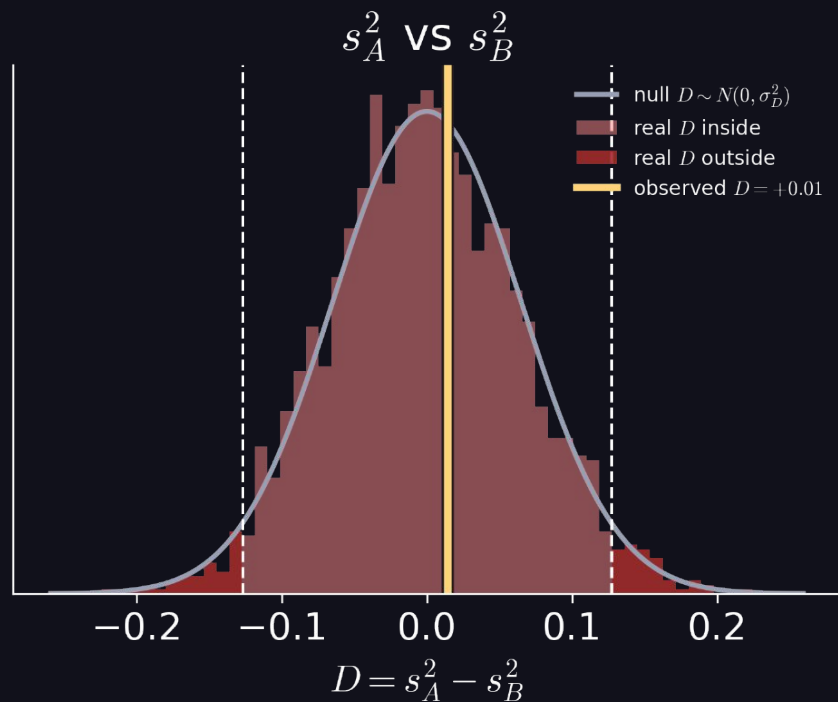


2. D for sample variances

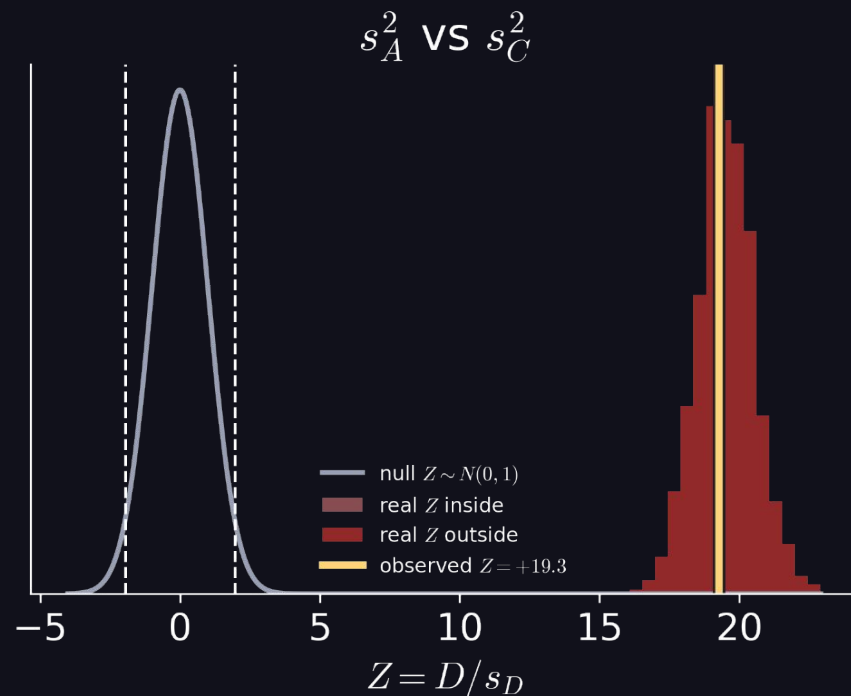
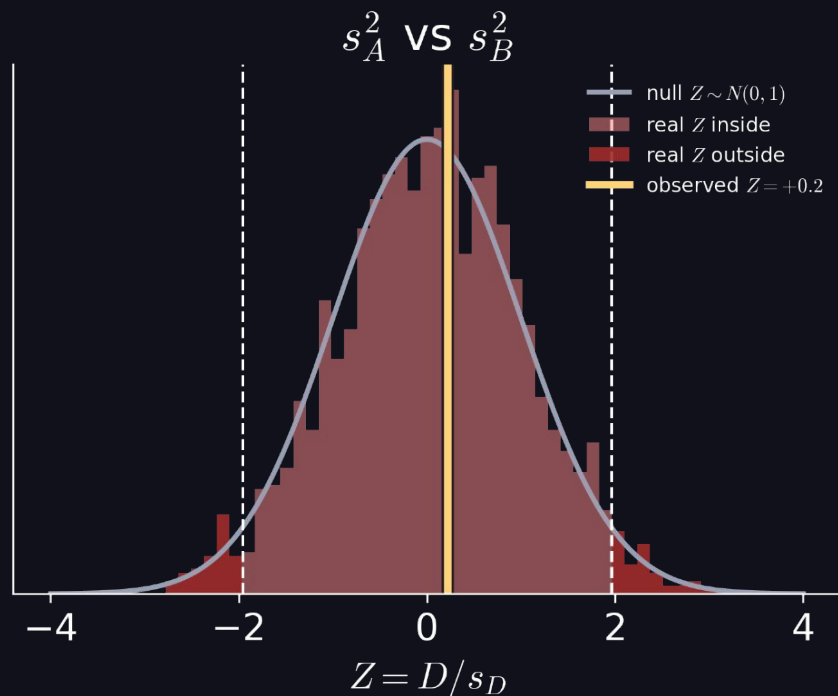
$$D = s_A^2 - s_B^2 \stackrel{\text{CLT}}{\approx} N(\sigma_A^2 - \sigma_B^2, \text{Var}(s_A^2) + \text{Var}(s_B^2))$$



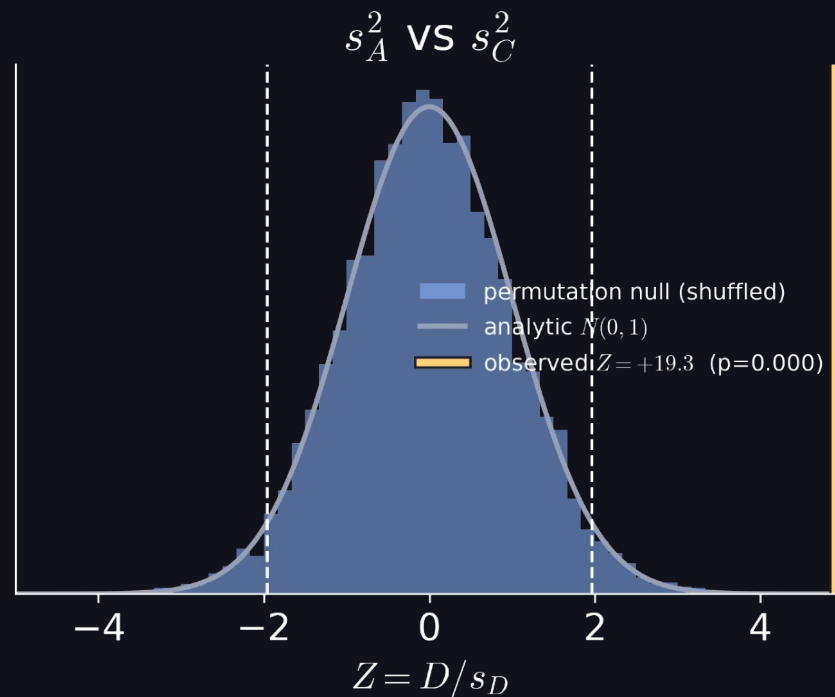
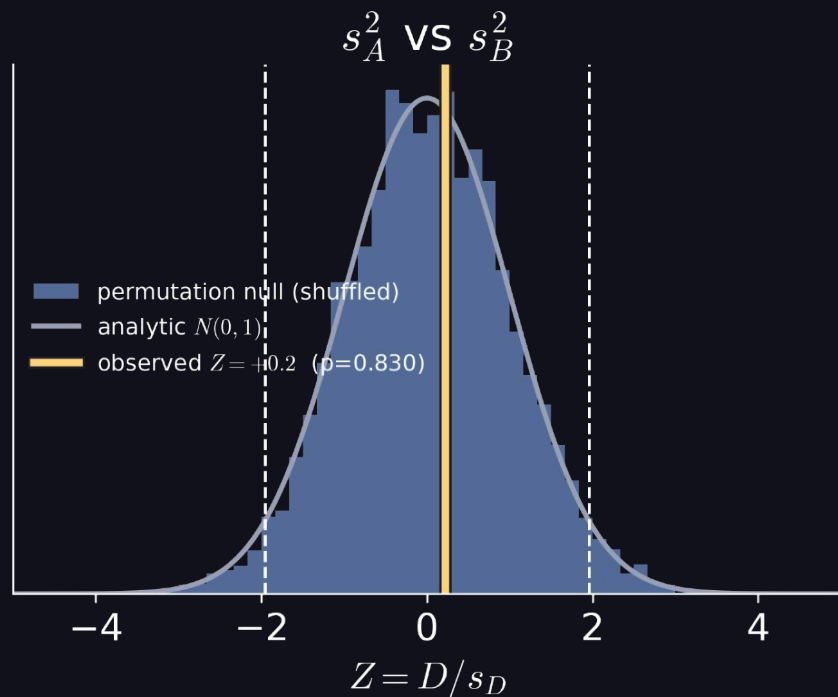
2. Null D for sample variances



2. Null Z for sample variances

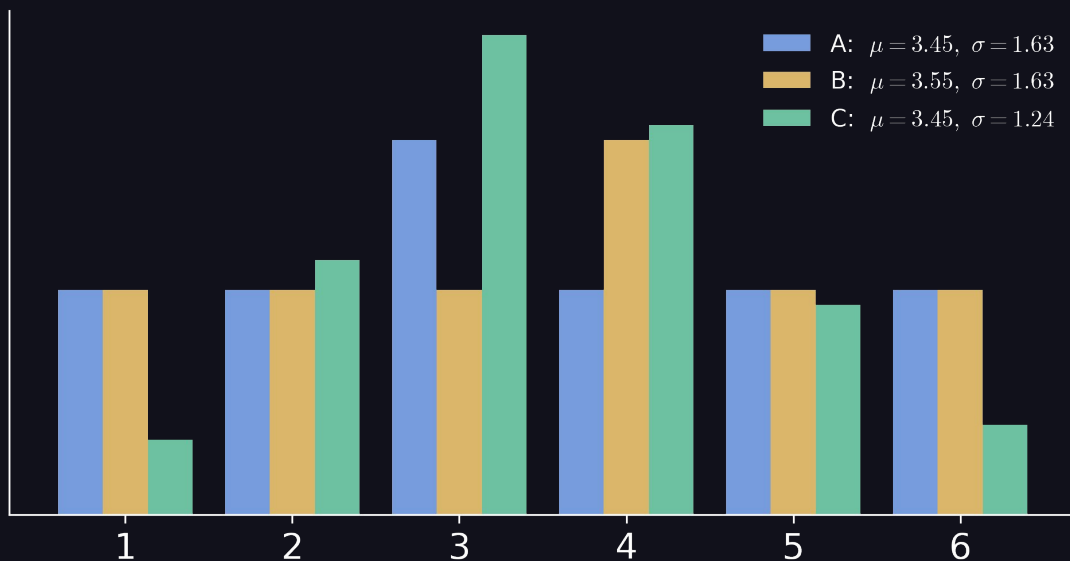


2. Permutation test for sample variances



2. What it taught me

To figure out what test you want to use you need to understand your null hypothesis.



2. Other test variants

- One/two tailed
 - Did the medicine take intended directed effect/any effect at all?
- Paired/unpaired
 - Can you match the datasets one-to-one or not?
- Rank
 - Ignores values, instead compares the orders

3. The paper



HOME | SUBM

New Results

 [Follow this preprint](#)

Widespread use of invalid statistical tests in biomedical machine learning

 Tianchu Zeng,  Hetu Li,  Shaoshi Zhang, Yan Quan Tan, Fang Tian,  Csaba Orban,  Lijun An,  Wanyu Che, Jingwen Cheng,  Joanna Su Xian Chong, Niousha Dehestani, Zijian Dong, Xin Li, Zhizhou Li,  Mervyn Jun Rui Lim, Yi Lin, Qinrui Ling, Zijie Ling,  Xi Zhi Low,  Sina Mansour L., Eric Kwun Kei Ng,  Thuan Tinh Nguyen,  Leon Qi Rong Ooi, Shreya Pande, Xing Qian, Jingxuan Ruan, Ziwen Wang, Yapei Xie,  Chen Zhang, Yichi Zhang,  Kaustubh Patil,  Linden Parkes,  Elvisha Dhamala,  Sidhant Chopra,  Andrew Zalesky,  Avram Holmes,  Simon Eickhoff,  Juan Helen Zhou,  Olivier Renaud,  Nico Dosenbach,  Konrad Kording, Danilo Bzdok,  Thomas E. Nichols,  B.T. Thomas Yeo

doi: <https://doi.org/10.64898/2026.05.17.724301>

3. Main idea for me

- When we apply stat tests to cross-validation results, we ignore that they are correlated
- This leads to underestimated variance (real variance is greater)
- This leads to more false positives
 - Null is rejected based on a wrong variance

3. How the paper suggests to fix it

- Split the dataset in halves
- Do CV on each half, get the differences in how models perform in each half/fold
- Average per fold -> get D_a and D_b
- Repeat the split several times -> get statistics on D_a and D_b
- See if D_a and D_b are the same or not.